

Program : Diploma in Biomedical Engineering	
Course Code : 6242B	Course Title: Rehabilitation Engineering
Semester : 6	Credits: 4
Course Category : Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester : 60

Course Objectives:

This course aims to

- Familiarize students with the combination of engineering technologies that can assist disabled individuals with mobility, communication, day-to-day activities for independent living.
- Impart knowledge on how to design and develop an active prosthesis.

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize rehabilitation fundamentals and rehabilitation concept for various disabilities.	13	Understanding
CO2	Describe engineering concepts in motor rehabilitation.	20	Understanding
CO3	Exemplify the design concepts of active prostheses and other electronic travel applications.	10	Understanding
CO4	Identify the sensory augmentation and substitution.	15	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize rehabilitation fundamentals and rehabilitation concept for various disabilities.		
M1.01	Describe the principles and basic methods used in rehabilitation engineering design	6	Understanding
M1.02	Identify the personal limitations and assistive technology requirement resulting from impairments and disabilities in the sensorial, motor and cognitive areas.	5	Understanding
M1.03	Describe engineering concepts in motor rehabilitation and sensory rehabilitation.	2	Understanding
Contents: Rehabilitation engineering - Definition - goals - Principles - Assistive technology device (definition and block diagram) and Assistive technology service (definition) - Personal and Social factors in assistive technology device design - design steps - intelligent technologies in rehabilitation. Impairments, disabilities and handicaps - Primary & secondary Disabilities - measurement and assessment - user's preferences, cognitive, physical abilities and limitations - rehabilitation team Rehabilitation - classification - motor rehabilitation - sensory rehabilitation			
CO2	Describe engineering concepts in motor rehabilitation.		
M2.01	Outline the design concepts of orthopedic orthotics	10	Understanding
M2.02	Describe the design and development of orthopedic prosthetics	6	Understanding
M2.03	Exemplify the Rehabs for mobility	4	Understanding
	Series Test - I	1	

Contents: Orthosis - Definition - General principles of Orthosis - biomechanics of orthosis - Classification of orthotics - functional, regional and specific conditions (give examples) - materials used - calipers -Ankle foot orthosis - Knee Ankle foot orthosis - Hip Knee Ankle foot orthosis - spinal orthosis - splints. Prosthetics - definition - classification (exoskeletal and endoskeletal) - components of a prosthesis - upper limb prosthesis - prosthesis for lower extremity. Mobility aids - functions - walking aids - crutches - wheel chairs - structure and parts of wheel chair - modified wheel chairs			
CO3	Exemplify the design concepts of active prostheses and other electronic travel applications		
M3.01	Describe the construction and working of externally powered and controlled orthoses and prosthesis	4	Understanding
M3.02	Exemplify the design and development of Intelligent prosthetic	6	Understanding
Contents: FES systems - principle - components - restoration of hand, restoration of standing and walking - HAS - hybrid assistive system for walking Active prosthetic design - principle - information flow (block diagram) - Human factors - man - machine interaction - Myoelectric Prosthesis - Intelligent Prosthetic Knee - Hierarchically Controlled Prosthetic Hand - Self-aligning Orthotic Knee Joint.			
CO4	Identify the sensory augmentation and substitution.		
M4.01	Describe visual substitution techniques.	5	Understanding
M4.02	Exemplify devices used for tactual auditory substitution	6	Understanding
M4.03	Explain tactual substitution systems.	4	Understanding
	Series Test - II	1	
Contents: Visual augmentation - Visual systems - Tactual vision substitution - Auditory vision substitution Auditory system - Auditory augmentation - Hearing Aids - Cochlear Implantation - Visual auditory substitution - Tactual auditory substitution Tactual system - Tactual augmentation - braille, optacon, optical character recognition.			

Text / Reference:

T/R	Book Title/Author
T1	Sunder S. - <i>Textbook of Rehabilitation</i> - Jaypee,2012
R1	Bronzino Joseph D. - <i>Handbook of Biomedical Engineering, second edition - Volume III</i> - CRC press,2010
R2	Cooper Rory A. - <i>An Introduction to Rehabilitation</i> - Taylor and Francis, London, 2012
R3	Horia - Nocholai, T. and Jain, L.C. - <i>Intelligent Systems and Technologies in Rehabilitation Engineering</i> - CRC Press (2001).
R4	Robinson C.J - <i>Rehabilitation engineering</i> - CRC Press (2006).

Online Resources:

Sl.No	Website Link
1	http://www.gpamdavadbme.hpage.com/
2	www.sciencedirect.com
3	www.mhrd.gov.in/e-content
4	http://www.gpgbiomedical.hpage.com/